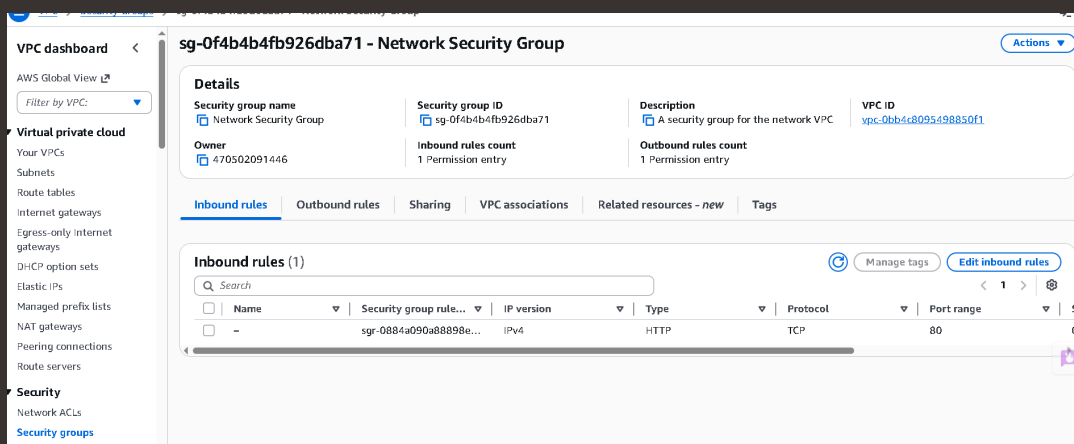




VPC Traffic Flow and Security



Lenah Mbatha





Route tables

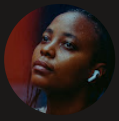
Route tables are tables containing rule or instructions that decide where the data in your network should go to

Routes tables are needed to make a subnet public because they control traffic where it is allowed to go

VPC > Route tables > rtb-0575498d08cd27118 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	local	-	No	CreateRoute
	Internet Gateway			
	igw-052a8b6c90792675f			



Route destination and target

Routes are defined by their destination and target, which mean ; 1. destination - the ip address range that the traffic wants to reach target- the road or the path that the traffic willhave to take to rach its destination

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of 0.0.0.0/0 and a target of igw-052a8b6c90792675f

VPC > Route tables > rtb-0373a9e0d8cd27118 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	local	-	No	CreateRoute
	Internet Gateway			
	igw-052a8b6c90792675f			

Add route

Remove



Security groups

Security groups are security checkpoint embedded in each resource in a vpc

Inbound vs Outbound rules

Inbound rules are rules that govern that data that enters your resources in your security group. I configured an inbound rule that allows users to access my webiste

Outbound rules are the rules that govern the data that your resourcescan send out. By default, my security group's outbound rule are allowed in my AWS account



VPC dashboard

Filter by VPC:

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

sg-0f4b4b4fb926dba71 - Network Security Group

Actions

Details

Security group name

Network Security Group

Security group ID

sg-0f4b4b4fb926dba71

Description

A security group for the network VPC

VPC ID

ypc-0bb4c8095498850f1

Owner

470502091446

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

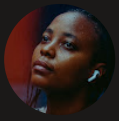
Inbound rules (1)

Manage tags

Edit inbound rules

Search

	Name	Security group rule...	IP version	Type	Protocol	Port range	
☐	-	sgr-0884a090a88898e...	IPv4	HTTP	TCP	80	

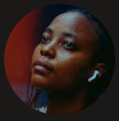


Network ACLs

Network ACLs are security checkpoints that makes sure only authorized data comes in and out of a subnet

Security groups vs. network ACLs

The difference between a security group and a network ACL is that in a security group, traffic rules apply at a resource level while in a network access controllist, traffic rules apply at a subnet level



Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow data to come in and out of a subnet, not unless configured otherwise

In contrast, a custom ACL's inbound and outbound rules are automatically set to deny until you set the rules to show which data packet traffic is allowed

The screenshot shows the AWS Management Console interface for Network ACLs. A green notification banner at the top states: "You have successfully updated subnet associations for acl-07afb01a98d669bff / Nextwork Network ACL." Below this, the "Network ACLs (1/3)" section displays a table of ACLs. The "Nextwork Network ..." ACL is selected, and its details are shown below, including the "Subnet associations (1)" table.

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound
-	acl-083adf952200567a3	3 Subnets	Yes	vpc-02012db0ae8df6c32	2 Inbound
-	acl-03132daeb21ecb4a0	-	Yes	vpc-0bb4c8095498850f1 / NetworkV...	2 Inbound
Nextwork Network ...	acl-07afb01a98d669bff	subnet-0d0da31a4dd96342f / Public 1	No	vpc-0bb4c8095498850f1 / NetworkV...	2 Inbound

Name	Subnet ID	Associated with	Availability Zone	IPv4 CIDR	IPv6 CIDR
Public 1	subnet-0d0da31a4dd9...	acl-07afb01a98d669bff / Nextwork N...	eun1-az1 (eu-north-1a)	10.0.0.0/24	-



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